Lesson 14 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

This Week: The Foreshadowed Feature

Since Lesson 11, several Teacher's Keys have deliberately flagged that hard-coding 5 enemies per wave and a fixed speed were 'deliberate simplifications' with 'a preview of Lesson 14's difficulty scaling.' This is that payoff lesson: enemy count and enemy speed both increase with the wave number, using the exact wave variable every engine has had since Lesson 11.

- ☐ Read the Handout's Level 1/2 split before class - Level 1 is enemy-count scaling only (a formula change, no new variables anywhere); Level 2 adds enemy-speed scaling (a new per-instance variable in engines that don't already have one).

Also This Week: Student's Choice

Level 3 this week is different from every prior lesson: instead of 2-3 small stretch ideas, students choose ONE feature from a menu of five capstone-level options (destructible shields, a power-up, a bonus enemy, formation-marching movement, or a title/game-over screen). These are intentionally lighter-touch than a full worked Level 1 - point students to the general shape and let them problem-solve using tools they already know.

- ☐ Skim the Level 3 menu before class so you can point students toward the option that best matches their engine and comfort level.

Self-Contained Lesson Assets

No new assets this week - difficulty scaling reuses existing enemy sprites and spawn logic. Any Level 3 stretch feature a student chooses may call for its own placeholder art; that's fine to improvise from existing lesson assets or simple shapes.

Continuity From Lesson 13

- ☐ Remind students to open their own [YourName]Game project from Lesson 13, not create a new one.
- ☐ If a student's Lesson 13 project is lost or broken, have a pre-built "Lesson 13 complete" starter available per engine (from the Lesson 13 Teacher's Keys) so they aren't blocked.
- ☐ 001 Game Creator (Windows demo laptop): confirm the two additional pre-placed enemy copies (enemy_6, enemy_7) needed for this lesson's count-scaling are added and named correctly before class - see this engine's Handout Setup section.

Engine Verification

- ☐ Confirm the demo laptop's Lesson 11-13 exercises (multiple waves, sound, hit reactions) still work for all 8 engines.
- ☐ Test a full 3-wave playthrough on every engine before class: wave 1 should feel like Lesson 13's game, and wave 3 should be noticeably faster and busier - but still fair and beatable.
- ☐ Confirm restarting still correctly resets to wave 1's original count/speed, not whatever wave 3 had scaled to.

Materials

- ☐ Print or queue digital copies of all 8 Student Handouts (or confirm students can access their engine's version electronically).
- ☐ Have the 8 Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies.

Known Gotchas This Week

- The most likely mistake: forgetting to reset enemies_remaining using the SAME per-wave count formula used in spawn_wave() - if these two places disagree, the win-check can fire early or never.
- A close second: forgetting that restart must return everything to wave 1's values, not leave speed/count stuck wherever they last scaled to.
- Difficulty scaling can make a game feel unfair if increments are too aggressive - if a cluster's wave 3 feels impossible rather than exciting, help them dial back the per-wave increment rather than pushing through.
- Since Level 3 this week is open-ended student choice, expect a wider range of finishing points than usual - that's fine; the goal is a good-faith attempt at ONE chosen feature, not a complete implementation.